

Question 1: Data Type Classifier

Write a function that accepts one value as a parameter.

The function should print whether the value is:

- Integer
- Float
- String
- Boolean
- List
- Tuple
- Dictionary
- Set

Use the `type()` function.

Question 2: Multiple Type Checker

Create a list containing different data types.

Without using loops, print the type of every element individually.

Question 3: Simple Data Validator

Accept input from the user.

Attempt to convert it into:

- Integer
- Float

If conversion fails, print an appropriate error message using `try...except`.

Question 4: Function Returning Different Data Types

Write four functions.

Each function should return a different data type:

- Integer
- Float
- String
- Boolean

Print the returned value and its type.

Question 5: Nested Dictionary Access

Create the following dictionary:

```
employee = {  
    "id":101,  
    "personal":{  
        "name":"John",  
        "age":30  
    },  
    "salary":55000  
}
```

Print:

- Employee ID
- Name
- Age
- Salary

using indexing with keys.

Question 6: List of Dictionaries

Create the following structure:

```
employees = [  
    {"name":"John","age":28},  
    {"name":"Alice","age":30},  
    {"name":"David","age":35}  
]
```

Print:

- Name of first employee
 - Age of second employee
 - Name of third employee
-

Question 7: Type Conversion Calculator

Accept two numbers from the user.

Store them as strings.

Convert them into integers and display:

- Addition
- Subtraction
- Multiplication

- Division
-

Question 8: Recursive Nested Access

Create the following variable:

```
data = [  
    10,  
    [  
        20,  
        [  
            30,  
            [  
                40  
            ]  
        ]  
    ]  
]
```

Using indexing only, print:

- 10
 - 20
 - 30
 - 40
-

Question 9: Dynamic Variable Inspector

Accept a value from the user.

Print:

- Original value
- Data type
- Length (only if it is a string)

Use an `if` statement to avoid errors.

Question 10: Mini Data Type Analyzer

Write a menu-driven program.

Menu:

1. Integer
2. Float
3. String

4. Boolean
5. Exit

Based on the user's choice:

- Accept a value of the selected type.
- Display the value.
- Display its data type.
- Perform one simple type conversion (for example, integer to float or string to integer where applicable).

Repeat the menu until the user chooses **Exit**.